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Use of surfactants to remove solvent-based inks from plastic films

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Abstract There is substantial economic and environmental incentive to remove the ink (deink) from heavily printed plastic films so that the plastic can be reused to produce clear films. In this study, polyethylene films printed with solvent-based ink were deinked using surfactants under a variety of conditions. Water without surfactant does not deink the solvent-based ink from plastic films over a pH range of 3 to 12. In common with earlier studies of water-based inks, it is found that solutions of cationic surfactants are the most effective for deinking of solvent-based ink but a pH of at least 11 is required. Presoaking of plastic film in aqueous solution of cationic surfactant increases the level of deinking. Limited studies performed with a pilot-size paper deinking apparatus on solvent-based ink removal indicates that the deinking of plastic film using surfactant solutions is technically feasible.

Keywords Deinking · Plastic film recycling · Solvent-based ink · Surfactant · Total color difference

Introduction

Recycling of plastic film from industrial and residential waste streams is gaining importance owing to the demand for raw materials, environmental concerns, and solid waste considerations. The fraction of plastic in USA municipal solid waste was estimated to be 9% by weight and 20% by volume; of this, 21–57% is plastic film [1, 2, 3]. Only about 3.5% of all plastics and 5% of plastic films are now recycled, and most research on plastic recycling has focused on rigid containers [1, 4].

In-plant, plastic film recycling is successful if the source is known and the scrap plastic is clean; however, most scrap films are heavily printed. Polymers produced from recycled printed film are not as valuable as virgin polymers because the residual ink colors repelletize resins and changes the physical properties of plastic. During reprocessing, residual ink also decomposes and produces gases that can crack pellets [5, 6]; therefore, the scrap plastic films are used to make low-value colored products [3]. However, there exist abundant sources of recyclable plastic films for scrap processing and great amounts of scrap plastic films are disposed off in landfills or are incinerated.

Effective film deinking technology would allow for the reuse of scrap plastic film by the primary fabricators as a raw material for producing clear films. Solvent-based deinking techniques such as the NOREC process have been developed and used in Europe [7]. However, these techniques are expensive, complicated, and produce hazardous waste at the end. For this study, we investigated the use of aqueous surfactant solutions for plastic film deinking since surfactants are more environmentally compatible [5, 6].

Polymers used for films include low-density polyethylene (LDPE), high-density polyethylene (HDPE), polyethylene terephthalate, ethylene vinyl acetate, poly(vinyl chloride), and polypropylene. Most plastic packaging is made of either LDPE, HDPE, or a mixture of LDPE and HDPE [8, 9,10].

The plastic film surface is almost always treated using one of several alternative surface treatment processes before printing to increase the surface energy of the plastic surface and enhance ink wettability. These surface treatments are flame treatment, corona discharge, plasma treatment, and UV treatment. Treatment with corona discharge is common and involves application of a high voltage (about 10,000 V) to an electrode which is positioned a short distance from the plastic film. The air between the electrode and the plastic is ionized, and this corona of ionized air oxidizes the surface [11].

Water-based or solvent-based ink is used for printing the plastic film surface with flexographic or gravure printing processes. The main ingredients of the inks are pigments, binders, carriers (vehicles), and additives. Organic and inorganic pigments give color and opacity to the ink and influence its fluidity. Binders, which are mostly low-molecular-weight polymeric resins, disperse the pigments and retain them on the plastic film surface after printing. The carrier (or vehicle) is a liquid, which provides fluidity for the ink and transfers the ink from the printing system to the substrate. Solvent-based ink uses solvent, solvent mixtures, or water-miscible solvent as the carrier [12]. Water-based ink uses water as the carrier, but might contain up to 20% alcohol. After application, the carrier should evaporate quickly and completely. Additives in the ink include waxes, surfactants, drying agents, and antioxidizing agents. Hence, the ink on a plastic film surface is a combination of pigments, binders, and additives. Water-based inks occupy a large and increasing fraction of plastic film inks since environmental regulations are restricting the percentage of volatile organic compounds from printing press emissions [13, 14, 15, 16]. However, more energy is required to dry water-based ink. Water-based ink also requires a higher surface energy of the film for good wet out and produces lower quality printing compared to solvent-based ink.

In our previous study, surfactants were determined to be effective in the removal of water-based ink from polyethylene film [5, 6]. Deinking, which involves the removal of various colors and layers of ink from the plastic film, can be considered as a laundering process. This process requires the detachment of the soils (ink), which adhere to the surface by VAN der Waals, electrical, and mechanical forces, and the dispersion of soil in a washing bath. Mechanical action, process chemistry, pH, time, and temperature are important in both plastic deinking and laundering of textiles [5, 6, 17].

In this paper, a discussion is given of the use of surfactants to deink solvent-based ink printing from both commercial plastic packaging film and from plastic films printed with one color (yellow, pink, red, green, gold, black, or violet). Removal of ink from solvent-based ink printed plastic films has been studied as a function of surfactant type, surfactant concentration, pH, agitation time, and presoaking. Scaling up of this solvent-based ink removal was conducted using a paper deinking apparatus.

Experimental

Materials

Sodium dodecyl sulfate (SDS) and hexadecyltrimethylammonium bromide (CTAB) were obtained from Sigma Chemical with a purity of greater than 99%. Dimethyldodecylamine oxide (DDAO) was provided by Fluka Chemical as a 30 wt% aqueous solution. Ethoxylated alcohol (AEO_5, trade name Witconol SN-70), ethoxylated amine ($AMEO_5$, trade name Varonic T-205), sodium dioctylsulfosuccinate (SDOSS, trade name Emcol 4500), and polyalkylene oxide modified polydimethylsiloxanes with molecular weights of 600 and 1,000 [AO-DMSi (600), trade name SILWET L-77 and AO-DMSi (1,000), trade name SILWET L-7607] were from Witco. All surfactants were used as received. The reagent grade calcium chloride from Fisher Scientific was used as received. Sodium silicate solution (SS, in a ratio of 3.3 SiO_2 to 1 Na_2O) was obtained from Sigma Chemical and was used as received. The water was double deionized.

A commercial plastic film (packaging of Kleenex UltraSoft) and plastic films printed with either yellow, pink, red, green, gold, black, or violet were supplied by Kimberly-Clark. These plastic films were composed of 70% LDPE and 30% HDPE. They were reverse printed with solvent-based ink by the flexographic printing process after surface treatment with corona discharge. The printed surface was then coated with white ink (overcoat) to give an opaque appearance and to protect the ink layers. Printed and clear parts of the plastic film were segregated for this study, and the clear part was used as a standard for comparison with deinked plastic film.

Methods

A backscatter image of the surface containing both clear and printed parts of the plastic film was taken by a scanning electron microscope (SEM, model JEOL JSM-880). A representative sample was mounted on a copper sample holder with double-sided carbon tape and coated with 60/40 gold/palladium in a Hummer VI sputtering system in order to make the sample surface conductive. After coating, a small dot of silver paint was placed on the edge of the sample to ensure conductivity. The same sample was also analyzed with an energy-dispersive X-ray spectrometer attached to the SEM in the spot mode.

All glassware was cleaned with a regular laboratory detergent and then acid-washed with Nochromix solution (Godax Laboratories) for at least 1 h. The surfactant solution was always prepared fresh with double deionized water for each of the experiments, and the pH level of the aqueous surfactant solution was adjusted immediately with Fisher-brand certified sodium hydroxide or hydrochloric acid before the deinking experiment was conducted. A representative sample (2.5 cm×2.5 cm) was cut from the printed part of the plastic film and soaked in a 20 mL solution (water or surfactant solution) with 20 irregularly shaped porcelain beads in 30-mL glass vials. The glass vial was then placed on a reciprocating-action Eberbach shaker for the desired period of agitation at high speed. After agitation, plastic film samples were taken out of the washing bath, rinsed with deionized water, and air dried. In addition, the deinking was scaled up to 2 L by using a Denver Laboratory model D-12 flotation machine with the attrition scrubber kit (Denver Equipment) [6]. After draining the washing bath, deinked plastic film strips were rinsed several times with water and dried in a vacuum oven overnight. Thirty plastic film strips (about 4 wt% of the deinked samples) were randomly picked up from the batch to analyze the level of deinking. All experiments were performed at room temperature and each experiment was performed at least twice.

Each deinked plastic sample was first evaluated visually and was then analyzed using a Hunterlab UltraScan™ spectrocolorimeter system (Hunter Associates Laboratory) in regular transmission mode. The Hunterlab UltraScan™ spectrocolorimeter system consists of an UltraScan optical sensor (SN-7834) with a sensor interface unit and an IBM computer that includes data-evaluation software. In the regular transmittance mode, the intensity of the light passing directly through the plastic film was measured wavelength by wavelength. The total color difference (DE*) between the standard (clear plastic film) and the sample (printed plastic film before and after deinking) was determined at the chosen standard illuminant (D65, representing daylight with a correlated color temperature of 6,500 K) and observer (10^0 standard observer). An average value from six measurements for each sample is reported. The lower the value of DE*, the better the deinking. More details about this procedure are given in Ref. [5].

Results and discussion

Deinking involves the removal of overcoat (white ink) and colored ink layers from plastic film. In Fig. 1, the backscatter images of the commercial plastic film taken by the SEM show the clear plastic film, ink layer, and overcoat. Energy-dispersive X-ray spectrograms of clear plastic film, colored ink, and overcoat from commercial plastic film are presented in Fig. 2. Energy-dispersive X-ray spectrograms of seven individual color printings, overcoat on these printings, and a clear part of the plastic film were also taken (not shown here). In these spectrograms, the detected copper is from the sample holder, while gold and palladium are from the sputter coat. Silicon was detected in all samples because silicon additives are used as flow-control agents, slip aids, defoamer, and hammer finish additives in ink formulations. Aluminum exists in the white overcoat and the gold ink layer, while titanium is only present in the overcoat layer. TiO_2 is a common pigment for white ink. Chlorine was detected in all ink layers except pink, black, and overcoat. Most of the organic pigments are halogenated. Calcium exists only in the pink ink layer.

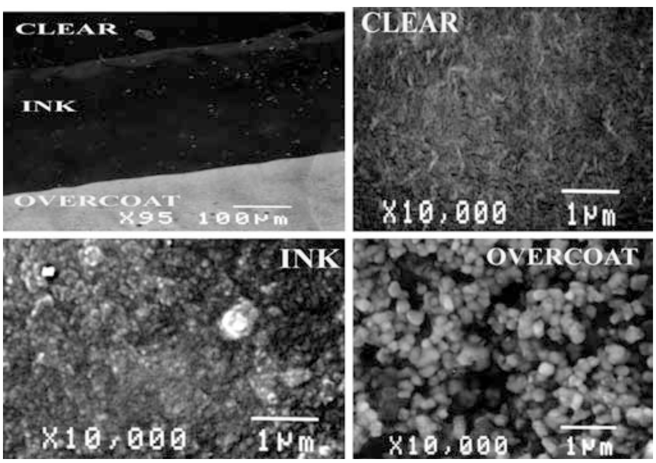

Fig. 1 Scanning electron microscope images of the commercial plastic film printed with solvent-based ink

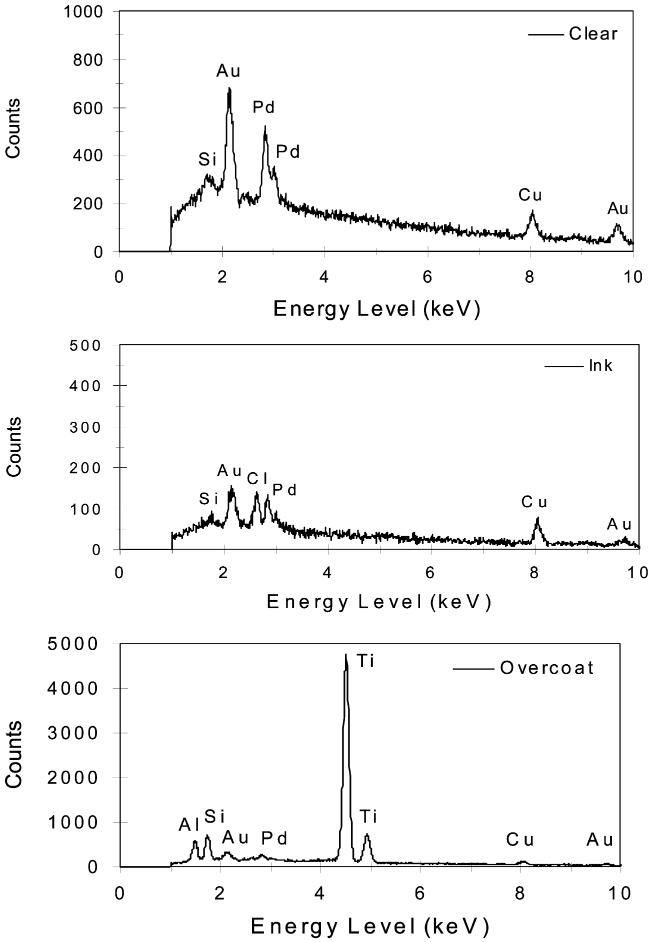

Fig. 2 Energy-dispersive X-ray spectrograms of clear film, ink, and overcoat layers (commercial plastic film)

Plastic film samples were agitated for 48 h with surfactant free basic solutions, but no detectable deinking occurred between pH 3 and 12. In contrast, surfactant-free basic solutions (pH 11 and higher) achieve partial deinking if the printing is with water-based ink probably because the binder is an acidic acrylate [5].

The commercial plastic packaging was treated with 1 wt% surfactant solutions at pH 10, 11, 11.5, and 12 with 2 h of agitation. The resulting DE* values are presented in Figs. 3, 4, 5, and 6. Complete deinking corresponds to a value of DE* of zero.

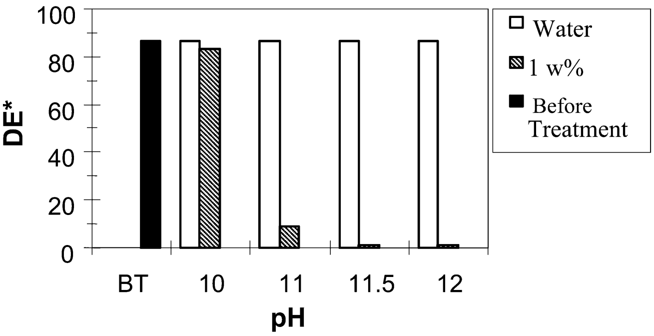

Fig. 3 Residual color of a commercial plastic film deinked with 1 wt% cationic surfactant with 2 h of agitation at various pH

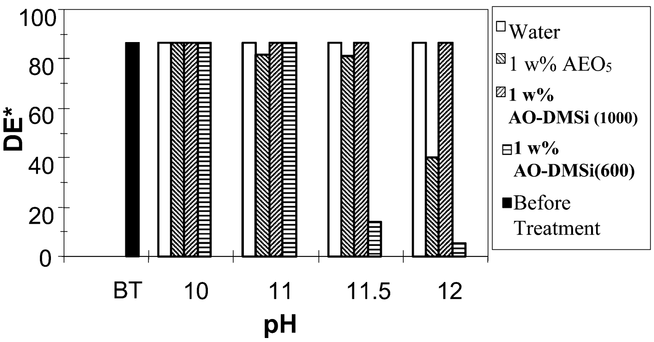

Fig. 4 Residual color of a commercial plastic film deinked with 1 wt% nonionic surfactants with 2 h of agitation at various pH. Ethoxylated alcohol (*AEO*$_5$), polyalkylene oxide modified polydimethylsiloxanes with molecular weights of 600 [*AO-DMSi (600)*] and 1,000 [*AO-DMSi (1,000)*]

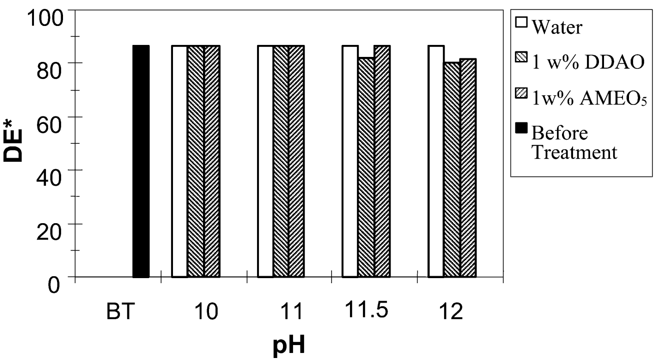

Fig. 5 Residual color of a commercial plastic film deinked with 1 wt% amphoteric surfactants with 2 h of agitation at various pH. Dimethyldodecylamine oxide (*DDAO*), ethoxylated amine (*AMEO*$_5$)

The cationic surfactant CTAB removes the solvent-based ink from the plastic film at pH 11, 11.5, and 12 (Fig. 3). The nonionic surfactants AO-DMSi (600) at pH 11.5 and 12, and AEO_5 at pH 12 (Fig. 4) partially remove the printing. Although the use of DDAO and

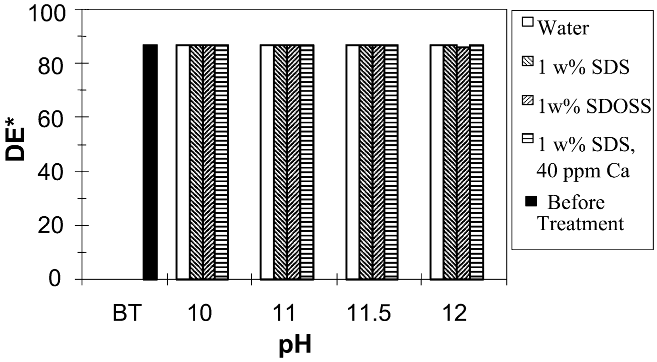

Fig. 6 Residual color of a commercial plastic film deinked with 1 wt% anionic surfactants in the absence and in the presence of 40 ppm calcium ions with 2 h of agitation at various pH. Sodium dodecyl sulfate (*SDS*), sodium dioctylsulfosuccinate (*SDOSS*)

$AMEO_5$ does not result in a significant change (Fig. 5), decolorization of the pink color on the plastic film at various levels was observed with the amphoteric surfactants DDAO and $AMEO_5$ at pH 12. The nonionic surfactant AEO_5 at pH 11.5 and the cationic surfactant CTAB at pH 10 also decolorized the pink color. The use of SDS, SDOSS, and SDS with Ca ions removes neither overcoat nor ink layers (Fig. 6). Sodium hydroxide is hypothesized to break up the print particle network by hydrolyzing the ester groups in the ink binder [18]. However, surfactants may be necessary to reduce the surface or interfacial tension at the air/water, ink/water, and plastic/water interface, and this can increase the wettability of the solution and allow alkali to penetrate between the ink particle and the plastic film.

The effects of agitation time and presoaking on deinking of the commercial plastic film were studied with CTAB at pH 11, 11.5, and 12. The soaking period was varied from 0 to 2 h, while the agitation time was varied from 15 to 60 min. As seen in Fig. 7, presoaking the plastic film strips in CTAB solution for 1 or 2 h at all

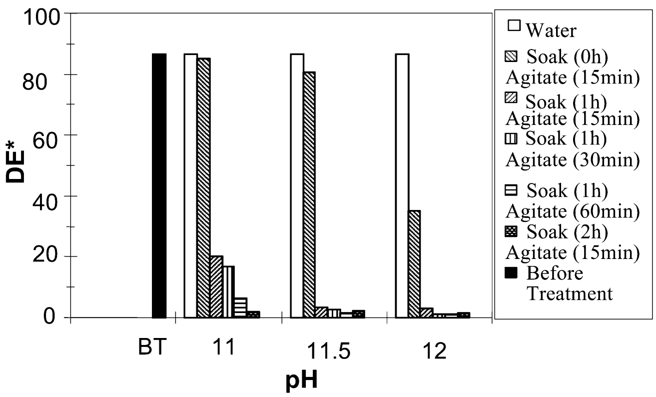

Fig. 7 Residual color of a commercial plastic film deinked with 1 wt% CTAB at various pH, soaking times, and agitation times

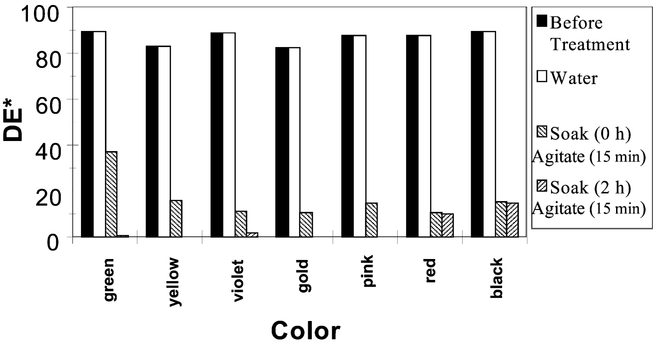

Fig. 8 Residual color of single-color printed plastic film deinked with 1 wt% hexadecyltrimethylammonium bromide (*CTAB*) with 15 min of agitation at pH 12 and various soaking times

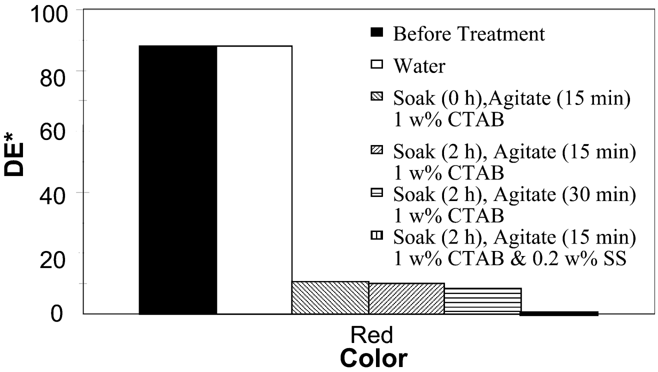

Fig. 9 Residual color of rose-colored ink printed plastic film deinked with 1 wt% CTAB or a mixture of 1 wt% CTAB and 0.2 wt% sodium silicate (*SS*) at pH 12 and at various soaking times, and agitation times

three pH levels significantly increases the deinking level at a given agitation time.

The deinking with CTAB (1 wt%) of plastic films printed with single colors at pH 12 with 15 min of agitation with and without 2 h of presoaking is shown in Fig. 8. Without presoaking, visual evaluation of the samples indicated that the white overcoat from all the samples except those with pink ink was totally removed, and a small portion of the ink stayed on the plastic film. However, the pink color was completely extracted in CTAB solutions. With presoaking, all colors except red and black are completely removed from the plastic film surface. Even increasing the agitation time to 30 min does not change the level of deinking for the red and black colors (Figs. 9, 10). However, as seen in Fig. 9, addition of 0.2 wt% SS into 1 wt% CTAB solution results in almost complete removal of red ink from the plastic film (DE* = 0.22). It has been suggested that silicate acts as a dispersing agent and creates a stable environment for peroxide in the paper deinking processes [19, 20]. On the other hand, the presence of SS in the surfactant solution does not improve the removal of black ink from the plastic film. The addition of AEO_5 (0.5 wt%) in CTAB (0.5 wt%) solution enhances the deinking of black ink (Fig. 10). These experiments show that the removal of black ink (which is carbon black) poses the greatest challenge in deinking. Our deinking studies with water-based ink printed plastic film also indicated that the level of deinking depends on the process parameters: pH, agitation time, agitation speed, presoaking, and temperature [5, 6].

Commercial plastic film pieces were soaked in 2 L 1 wt% CTAB solution at pH 12 at a plastic-to-solution ratio (consistency) of 0.5% for 2 h. After soaking, agitation was applied at 1,000 rpm by using a Denver flotation cell as a scrubber. During agitation, foaming of the surfactant solution occurred. After deinking, the weight of the plastic film strips decreased by about 9.5%. DE* was measured to be 1.35 ± 0.12 from 30 samples with a 90% confidence interval and is presented in Fig. 11. As seen in Fig. 11, increasing the plastic-to-solution ratio (consistency) from 0.1 to 0.5% and the volume from 20 mL to 2 L does not affect the deinking level. Plastic film strips printed with solvent-based ink are uniformly and almost completely deinked under these conditions.

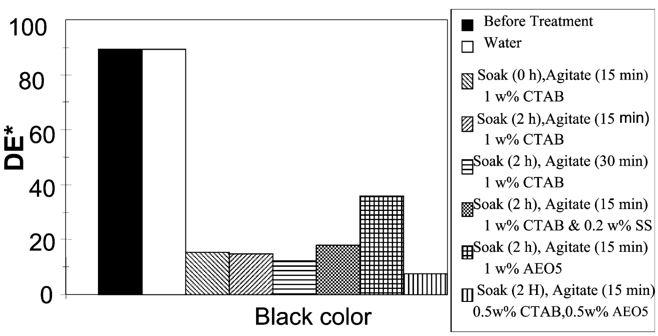

Fig. 10 Residual color of black ink printed plastic film deinked with 1 wt% CTAB, 1 wt% AEO_5, a mixture of 1 wt% CTAB and 0.2 wt% SS, and a mixture of 0.5 wt% CTAB and 0.5 wt% AEO_5 at pH 12 and at various soaking and agitation times

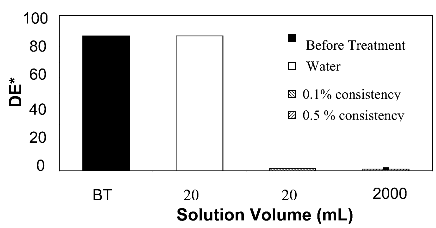

Fig. 11 Residual color of a commercial plastic film deinked with 1 wt% CTAB at pH 12 and various solution volumes with 2-h soaking and 15-min agitation

Conclusion

Cationic surfactant is the most effective surfactant for the removal of solvent-based ink from plastic film. The process pH is required to be adjusted to 11 or above for complete deinking. All colors except red and black inks can be completely removed from the plastic film by presoaking in a 1 wt% cationic surfactant solution at room temperature at pH 12. The addition of SS to the cationic surfactant completely removes red ink. Substantial, but not complete deinking of carbon black from plastic film was observed when plastic film was deinked with a mixture of nonionic and cationic surfactants.

In general, deinking of solvent-based inks using aqueous solutions of surfactants requires severe conditions and a more restricted choice of surfactants than water-based inks. The process conditions and the selection of surfactants and additives are affected by the ink type and color. The use of aqueous solutions of surfactants rather than organic solvents to deink plastics should lead to major environmental benefits.

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